

SIOC 251 Homework 2: Simulating the Radiative Forcing of Well-Mixed Gases in the LW

In this exercise you will create a longwave radiative transfer model of monochromatic flux in order to investigate radiative forcing of a well-mixed gas. In a later homework we add more complexity to this model, so be sure to document your code.

Part I – Weighting Function

Let's assume we have some well-mixed gas in the atmosphere such that the extinction coefficient β_e can be expressed as

$$\beta_e = C \exp(-z/H)$$

where C is a constant relating gas abundance to extinction, z is height, and H is atmospheric scale height. Using this expression for β_e :

1. Derive expressions for $\tau(z)$ and τ^* .
2. Derive an expression for $t_F(z, \infty)$.
3. Derive an expression for $W_F^\uparrow(z)$

Next, define C as

$$C = f \times 1.25 \times 10^{-4} (\text{m}^{-1})$$

where f is a scaling parameter that is equivalent to changing the concentration of this well-mixed gas in the atmosphere.

4. How does $W_F^\uparrow(z)$ change for $f = 0.001, 0.01, 0.1, 1, 10, 100$ (i.e., plot $W(z)$ for all values of f)?
5. How would your changes in $W_F^\uparrow(z)$ affect the TOA outgoing LW radiation?

Part 2 – TOA outgoing LW radiation

This exercise will be easier to complete if you calculate the various monochromatic fluxes for continuous values of f , and not just the six values defined in Part 1.

Using the results from Part 1:

1. Derive an expression for the monochromatic upwelling flux at TOA $F_{\lambda}^{\uparrow}(\infty)$.
2. Write an expression for the monochromatic net flux at the TOA, $F_{n,\lambda}(\infty)$ (recall that we define the net flux as $F_n = F^{\uparrow} - F^{\downarrow}$)?

Define the direct radiative forcing of this gas at some wavelength as

$$\Delta F = F_{n,\lambda}(\infty; f) - F_{n,\lambda}(\infty; f = 0),$$

which is to say that the radiative forcing is the net flux for some gas concentration $C(f)$ minus the net flux for $C = 0$. Now, assume that $\lambda = 15\mu\text{m}$, the diffusivity parameter $r = 5/3$, and temperature profile $T(z)$ is given by

$$T(z) = \begin{cases} T(z=0) - \Gamma z, & \text{if } z \leq 20\text{km} \\ 200, & \text{otherwise} \end{cases}$$

where $T(z=0) = 300\text{ K}$ and $\Gamma = 5\text{ K km}^{-1}$.

3. Calculate $F_{n,\lambda}(\infty)$ for all values of f
4. Calculate and plot the TOA radiative forcing $\Delta F(\infty)$ as a function of f .
5. Explain the shape of $\Delta F(\infty)$ in terms of changes in $W_F^{\uparrow}(z)$ and $T(z)$.

Repeat the above exercise, but for a slightly more realistic $T(z)$,

$$T(z) = \begin{cases} T(z=0) - \Gamma z, & \text{if } z \leq 20\text{km} \\ 200 + \Gamma_s z, & \text{if } z > 20\text{km} \& z \leq 50\text{km} \\ 260, & \text{otherwise} \end{cases}$$

again for $T(z=0) = 300\text{ K}$.

6. Answer questions 3–5 for this temperature profile.

Again... Repeat the above exercise, but for an isothermal atmosphere

$$T(z) = 300 \text{ K}$$

7. Answer questions 3–5 for this temperature profile.

You Are Done!

Write up your homework into a single PDF file and upload to Canvas.